A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Mineral resources (the lithosphere) - Set 1

Mark scheme

AQA spec 3.2.3; The physical environment; textbook pp. 130-147

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1** Igneous intrusions (batholiths) of magma cool within the crust, heating groundwater to produce superheated, pressurised water **1** The hot water dissolves minerals from the surrounding rock and the mineral-rich solutions travel along fissures / cracks away from the batholith **1** As the solutions cool the dissolved minerals crystallise in order of solubility (least soluble first) - fractional crystallisation **1** So different metals are separated and concentrated in veins, eg tin, copper, lead, silver, gold, arsenic (SW England batholith) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Dissolved iron compounds in ancient seas were oxidised by the oxygen released by early photosynthetic organisms (2.5-1.8 billion years ago) **1** Iron oxides are insoluble so they were deposited as sediments (haematite, magnetite) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Flowing water carries solid particles – the ability to carry them depends on the water velocity and the density of the solids **1** As velocity falls, dense minerals (gold, tin ore) are deposited first while lighter material is carried further, concentrating the heavy minerals in particular places **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Limestone / chalk from the shells of marine organisms **1** Coal from terrestrial vegetation **1** Crude oil and natural gas from marine organisms (plankton) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 132-134; AQA spec 3.2.3.2.	
02.1	1 mark for the point below. **1** As the (linear) purity / grade of a mineral deposit decreases, the amount of the mineral present increases exponentially / logarithmically **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
02.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 2400 / 40 = 60 times **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 3 points below. **1 Reserves:** the amount that can be exploited now, economically, with existing technology (deposits above the cut-off ore grade) **1 Resource:** all the material that could be exploited technically and economically now or in the future (with realistic price rises or new technology), including deep, low-grade or protected deposits **1** The resource is finite but the reserves change: they rise if prices rise or technology improves, and fall if prices fall **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Lowering the cut-off ore grade includes deposits of lower purity in the reserves **1** Because the amount of material rises exponentially as purity falls, a small drop in COOG adds a very large mass (eg 0.5-1.0% has 4x the copper of 1.0-2.0%) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.5	**1 mark each** 1 mark for each of the 3 points below. **1** The total quantity (stock / resource base) is far greater than could ever be used **1** But most is at purities too low or in locations too inaccessible for current economic exploitation **1** The challenge is developing technologies (low-energy extraction such as bioleaching, phytomining; better exploration) that make low-grade deposits exploitable – energy needs rise rapidly as ore grade falls **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 135-136; AQA spec 3.2.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 4 points below. **1 Gravimetry:** gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) **1** Magnetometry **1 Seismic survey:** sound / shock waves from controlled explosions or a seismic vibrator – echoes recorded by geophones give the depth, density and shape of rock strata **1** Resistivity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** Trial drilling is the most expensive technique per site **1** Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites **1** Drilling is then targeted where deposits are likely – it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Different minerals emit / reflect infrared at different wavelengths **1** Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
03.4	1 mark for the point below. **1** Results are available immediately in the field, saving time waiting for laboratory analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 137; AQA spec 3.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** The lowest ore purity that can be mined economically with existing technology **1** Deposit X (0.9% is above 0.4%; Y at 0.35% is part of the resource, not the reserves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.2	**1 mark each** 1 mark for each of the 6 points below. **1** X has higher grade so less rock mined, less spoil, less energy and pollution per tonne of copper **1 But X is deep:** costs rise more than proportionally with depth (sides cannot be vertical so much more rock removed; more water inflow and pumping) **1** X has hard overburden requiring blasting (cost, noise, dust) and high rainfall increasing drainage costs **1** X is far from transport infrastructure so transport costs are high **1** Y is shallow with loose overburden (cheap to remove but landslide risk so gentler slopes / larger land take), near a railway, low rainfall **1** Y's grade is below the COOG so it is not viable now but would become viable if copper prices rose or extraction technology improved (eg bioleaching) – both sulfides risk acid mine drainage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
04.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Higher market value / price of the metal **1** Improved extraction technology **1** Lower energy costs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 138-139; AQA spec 3.2.3.5.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Minerals can only be exploited where they were deposited (no choice of location), unlike housing, roads or ports **1** Land take is larger than the mine void: buildings, access roads, overburden and spoil heaps, buffer zones – open-cast mines take far more land than deep mines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 5 points below. **1 Dust:** water sprays on roads / blasting areas make particles heavier and clump so they settle **1 Noise:** baffle mounds / embankments absorb and deflect – blasting at set, predictable daytime hours – quieter machinery **1 Turbid water:** sedimentation lagoons where water stands until solids settle before discharge **1 Leachate:** collect drainage water and pass through crushed limestone to neutralise acid and immobilise toxic metals **1 Amenity:** landscaping and tree planting to screen the mine – restoration plans agreed as a condition of planning permission **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Leachate** → polluted liquid draining out of waste	5
05.3	**1 mark each** 1 mark for each of the 3 points below. **1** Landscaping to reduce gradients, compaction, drainage to prevent waterlogging, adding soil / nutrients and planting vegetation whose roots bind the surface **1** At Aberfan a waterlogged coal-mine spoil heap became unstable and slid into the village **1** Killing 144 people, mainly children in the primary school **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogging** → soil pores filled with water so there is no air	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 139-142; AQA spec 3.2.3.6.	
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** The void fills with water naturally – wetlands are uncommon / valuable habitats in most regions **1** The new habitat may have greater wildlife value than the farmland lost – low cost compared with filling and levelling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1 Housing:** ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) **1 Farmland:** landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	1 mark for the point below. **1** Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost – transplanted plants often fail **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	
07.1	**1 mark each** 1 mark for each of the 5 points below. **1** Acidophilic bacteria (eg Acidithiobacillus) oxidise sulfide ores, producing sulfuric acid which dissolves the metal ions **1** Leachate (rainwater / recirculated solution) percolating through the spoil heap is collected – recirculation increases the metal concentration **1** Copper is recovered from solution by electrolysis or by displacement with scrap iron (iron is more reactive so copper is deposited as solid metal) **1** Smelting needs chemical reduction at high temperatures – energy needed rises rapidly as ore grade falls, so low-grade ores are uneconomic **1** Bioleaching works at low temperatures with low energy inputs, so it can exploit low-grade deposits and wastes (increasing reserves) - mimics natural low-temperature processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **Leachate** → polluted liquid draining out of waste **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	5
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** Plants (hyperaccumulators) absorb metal ions from soil / water and concentrate them in their leaves **1** The vegetation is harvested and incinerated **1** The concentrated metal in the ash is dissolved in acid and separated by electrolysis **OR** decontaminates polluted sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Uranium ions dissolved in seawater adsorb onto the surface of synthetic or natural polymers (lignin from wood, chitin from shrimp shells) **1** The polymers are collected and the uranium recovered, avoiding mining / high-energy extraction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 143-145; AQA spec 3.2.3.7.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** Metal-rich lumps 5-10 cm across on the deep seabed (4000-5000 m), containing about 30% manganese plus nickel, copper, cobalt, iron, titanium **1** Exploitation would be very expensive (remotely operated collectors, pumping to the surface) / no viable technology yet **1** International agreement on ownership of the seabed and its resources is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Collectors disturb the seabed and kill benthic organisms living on it **1** Separating nodules from sediment creates plumes that increase turbidity with unknown consequences **1** Re-deposition of sediment covers and kills filter feeders and other benthic organisms – slow recovery in the deep sea **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 4 points below. **1** Used in small amounts in many essential technologies (smartphones, hybrid vehicles, wind turbines, fibre optics, LEDs) **1** Exploitable land deposits are limited and mainly in China – chemical separation is complex and produces large amounts of toxic waste **1 Bacterial adsorption:** dissolved rare earth ions adsorb onto the cell surfaces of bacteria (Bacillus subtilis) and are washed off at specific pHs to separate them **1** Deep sea sediments contain relatively high concentrations but no viable extraction method yet **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 145-146; AQA spec 3.2.3.7.	

Do not write outside the box